

? Hackerspeak: Jargon Explained

Intrusion Detection: The processes involved in detecting inappropriate, incorrect or anomalous activity happening in a network.

Intrusion Detection Systems (IDS) are employed to determine if a computer network or server has fallen prey to an unauthorised intrusion.

Honeypots: Closely monitored network decoys that help distract intruders from attacking more valuable machines on a network and causing damage.

Honeypots ring the early warning bells about an attack and help study exploitation trends. Deployment of honeypots usually does not affect

critical network services or applications.

Challenge: A virtual system designed by programmers with various hacking-related goals/problems to be solved

Exploit: An application designed to penetrate known vulnerabilities in a network or software

Sniffer: A tool that captures passwords and other data while in transit within the computer or over a network

Social engineering: The practice of conning people into revealing sensitive

data on a computer system, often on the Internet. In other words, exploiting the weakest link—more often than not, users

Vulnerability scanner: A tool that scans computers on a network for known weaknesses. Port scanners look for “open ports” to gain entry into the network.

Man in the middle attack (MTM): Here, an attacker reads and modifies messages between two parties without letting either party know that their link has been compromised.

The attacker normally observes and intercepts messages that travel to and from the victims' PCs.

rity engineers normally protect the critical resources of a company—server side applications and Web sites.”

Hierarchies In Cyberia

At the bottom of the hacker hierarchy is a ‘script kiddie’—a malicious hacker with little or no skill. These guys can cause a lot of damage with their shoot-in-the-dark methods. A ‘white hat’ or ‘sneaker’, who also breaks into networks or seeks vulnerabilities, does so for ‘altruistic’ reasons. ‘White hats’ usually report bugs or vulnerabilities to the vendors.

‘Black hats’ or ‘Crackers’, on the other hand, are on the lookout for vulnerabilities in networks and software, and enjoy destroying vulnerable networks.

Top-notch hackers—usually referred to as ‘Wizards’ or ‘Gurus’—have been associated with some of the biggest names in the software business. The best-known amongst these are Richard Stallman, founder of the Free Software Movement and GNU projects, which nurtured the development of the GNU/Linux operating system (OS), and Linus Torvalds, who developed the Linux kernel in 1991.

Hackers get to match wits at events such as DEFCON (www.defcon.org) dubbed the “largest underground hacking event in the world”, the Black Hat conference (www.blackhat.com)—a melting pot of hackers, security experts, government officials and network administrators, and COMDEX (Communications and Data Processing Exposition)—a hardware and software tradeshow.

Enemy At The Gate

Vulnerabilities abound in networks. Security professionals are privy to this knowledge, especially during audits. They have no reason not to distribute this knowledge for profit or even exploit the vulnerability.

According to Puri, some of the most potent hacking attacks include distributed denial of service; idle scanning—exploiting the bugs on a server running on the network and then taking control of the rest of the infrastructure; sniffer attacks—where a sniffer is installed on one of

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Kartikeya Puri,
MIEL eSecurity

the servers and compromises the information flow; attacks on the border routers/ switches; overflows (buffer/heap/stack-based) in server software; and social engineering attacks.

Once a network has been compromised, the intruder gains ‘Administrator’ privileges, and using certain tools (‘root kits’) disables security auditing. At the end of their stay, using the same tools, the intruders simply turn on auditing again.

Says Prakash (name changed), a cracker-turned-security engineer from Bangalore, “Most networks cannot guarantee 100 per cent security. Online banking systems in India, for example, are vulnerable to attacks. But it involves a lot of work—traffic to the site has to be sniffed, data has to be encrypted, the network scanned for vulnerabilities, firewalls disabled, and so on. But a break-in is certainly possible.”

Dancing With The Devil

For every successful ethical hacker or systems administrator, there is a hotshot cracker waiting to get the better of him. Although there is big money involved, the cat-and-mouse game between security experts and malicious hackers is also a battle of extremely creative minds. Hackers are constantly upgrading their knowledge, while virus writers continue to unleash their creativity on the Web.

Genuine hackers “seek to create, nurture and contribute to the evolution of more efficient computer networks”.

“Ethical hacking is subjective; its interpretation varies—I could enter a network and snoop around without causing any significant harm. Does that make me any less ethical?”, asks Puri.

Yet, he confesses it is hard to stay ‘ethical’ when one has knowledge and access to sensitive and potentially lucrative information.

Thomas Anderson or Neo, from The Matrix, a “program writer for a respectable software company... and a hacker guilty of virtually every computer crime we have a law for...”, is merely a stereotype. Not all hackers are malicious and not all ethical hackers are saints. But which hat would you rather wear? ■

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